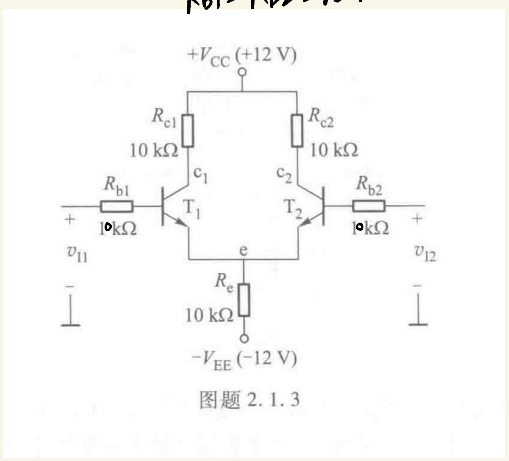
2.1.3 $R_{b1}=R_{b2}=10\ k\Omega$



图题 2.1.3

(1) $I_{CQ1}=I_{CQ2}=\frac{V_{EQ}-V_{BQ}}{2R_e+\frac{R_{b1}}{1+\beta}}\approx 0.56\ mA$ $V_{CEQ}=7.1\ V$

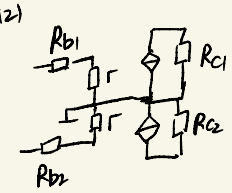
$V_{CE1Q}=V_{CC}-V_E=12.7\ V$

T_1, T_2 静态电流不变

T_2 电压减

T_1 电压变大

(2)

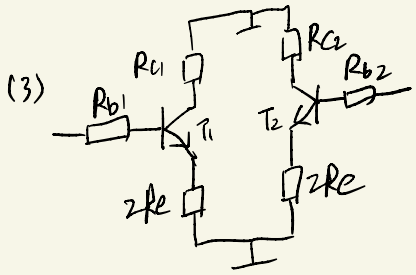


$r_{be}=r_{bb'}+(1+\beta)\frac{26mV}{I_{CQ}}=4.89\ k\Omega$

$R_{id}=2(R_{b1}+r_{be})=29.8\ k\Omega$

$\dot{A}_{vd2}=\frac{1}{2}\dot{A}_{vd}=\beta R_c/2(R_{b1}+r_{be})\approx 33.6$

(3)



$r_{be}=4.89\ k\Omega$

$\dot{A}_{vc2}=\frac{-\beta i_B R_c}{i_B r_{be}+(1+\beta)i_B 2R_e}=\frac{-\beta R_c}{r_{be}+2(1+\beta)R_e}=-0.5$

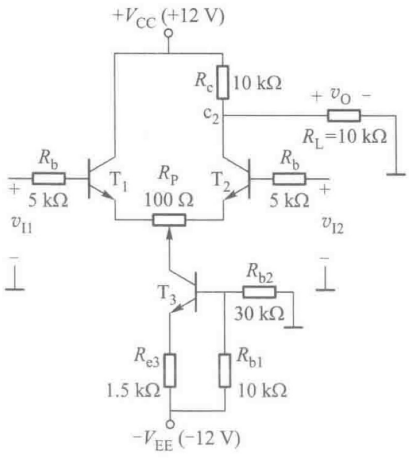
$K_{CMR}=20\lg\left|\frac{A_{vd}}{A_{vc}}\right|=36.5\ dB$

(4) $\Delta V_{c2}=V_{cd}+V_{ce}=A_{vd2}(V_{I1}-V_{I2})+A_{vc2}\frac{V_{I1}+V_{I2}}{2}=286\ mV$

$V_{c2}=V_{cc}-I_{cq2}R_2+\Delta V_{R_2}\approx 6.686\ V$

$V_E=V_{Ic}-V_{BE}=-0.6\ V$

2.1.6



图题 2.1.6

(1) $IR_{b2}+0.7+\beta I_{B3}R_{e3}=V_{EE}$ $I_{B3}=0.028\ mA$

$IR_{b2}+(I-I_B)R_{b1}=V_{EE}$ $I_{CQ3}=\beta I_{B3}=1.4\ mA$

$I_{CQ1}=I_{CQ2}=\frac{I_{CQ3}}{2}=0.7\ mA$

(2) $\dot{A}_{vd2}=-\frac{1}{2}\dot{A}_{vd}=\beta(R_c/\!/R_c)/2[R_b+r_{be}+(1+\beta)\frac{1}{2}R_p]\approx 13.1$

$R_{id}=2[R_b+r_{be}+(1+\beta)\frac{R_p}{2}]=19.1\ k\Omega$

$R_o=R_c=10\ k\Omega$

(3) $\dot{A}_{vc2}=\frac{-\beta(R_c/\!/R_c)}{R_b+r_{be}+(1+\beta)(R_p+2R_{e3})}\approx -0.003$

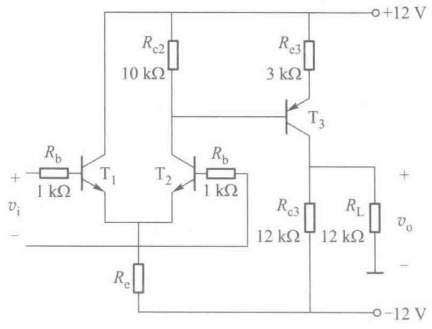
$K_{CMR}=20\lg\left|\frac{A_{vd2}}{A_{vc2}}\right|=72.8\ dB$

(4) $\Delta V_o=(V_{I1}-V_{I2})A_{vd2}+\frac{V_{I1}+V_{I2}}{2}A_{vc2}\approx V_{I1}A_{vd2}=0.262\ V$

$V_o=V_{oQ}+\Delta V_o=2.762\ V$

2.1.12

的输入电阻 R_i 和输出电阻 R_o。



图题 2.1.12

(1) $I_{CQ3}=\frac{12V}{R_{c3}}=1mA$

$I_{CQ2}'=(I_{CQ3}R_{e3}+V_{BE})/R_{c2}=0.37mA$

$12V=\frac{I_{CQ2}}{\beta}R_b+V_{BE}+2I_{CQ2}R_e$ $\therefore R_e=14.5k\Omega$

(2) 双端单端转换输出

$\dot{A}_{V_1}=-\frac{1}{2}\dot{A}_{od}=\frac{\beta(R_{c2}\parallel R_{i2})}{2(r_b+r_{be1})}=51$ $R_{i2}=r_{be3}+(1+\beta)R_{e3}=154.5k\Omega$

$r_{be1}=3.6k\Omega$ $r_{be3}=1.5k\Omega$

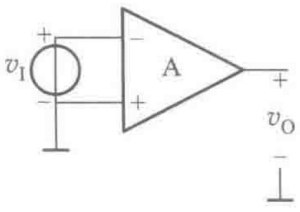
$\dot{A}_{V_2}=\frac{-\beta(R_{c3}\parallel R_L)}{r_{be3}+(1+\beta)R_{e3}}=-1.94$

$\dot{A}_{vd}=\dot{A}_{v_1}\cdot\dot{A}_{v_2}=-99$

(3) $R_i=2(R_3+r_{be1})=9.2k\Omega$

$R_o=R_{c3}=12k\Omega$

2.1.16



图题 2.1.16

(1) $V_i=1mV$ $V_o=-10V$

(2) $V_i=1.5mV$ $V_o=-10V$

(3) $V_o=-10V$，不能正常放大

2.1.17

(1) ✗

(2) ✓

(3) ✗

(4) ✓

(5) ✓